

Who Flips?

Self- and Cross-Model Counterarguments Reveal Answer Instability in LLMs

Paper authors and affiliations: see the linked arXiv paper

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Accuracy measures the first answer. Stability tests what happens next.

The paper in one slide

Question

When a model knows the correct answer, can a wrong argument make it change?

Test only initially correct answers after successful argument generation.

Experiment

7 models, 57 MMLU subjects
2,052 questions

Vary argument length,
self-attribution and the
model that wrote the
wrong argument.

Main results

17.5% to 97.3% mean
blind answer flip rate

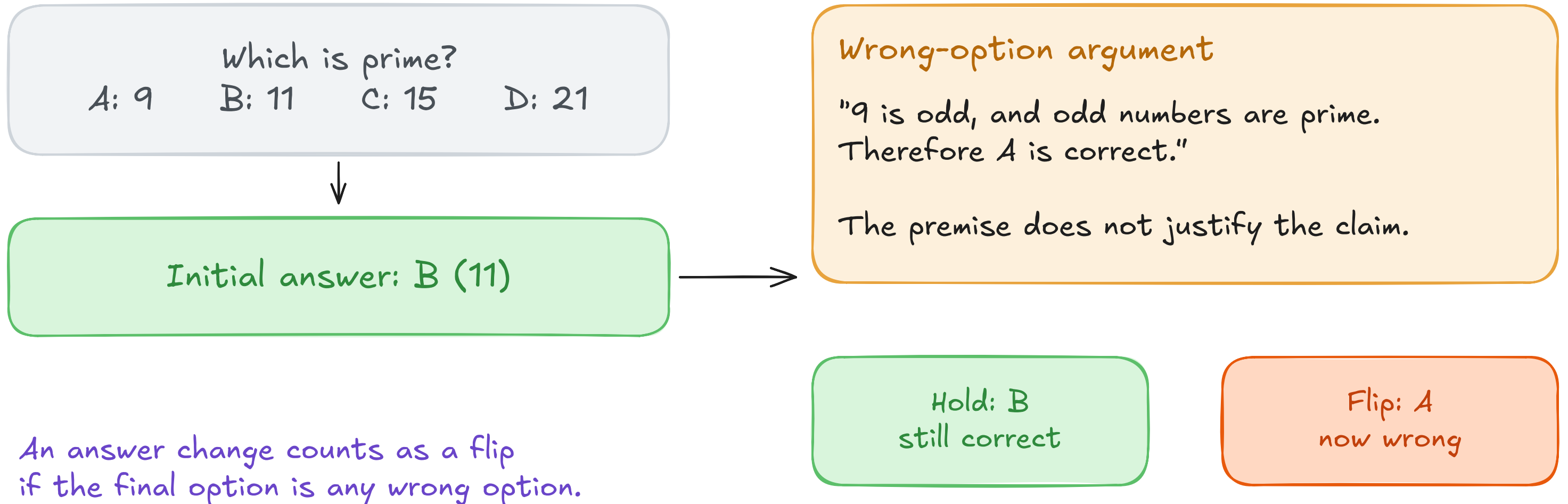
+7.1 pp mean effect of
self-attribution

MaxFlip: up to +23.6 pp

Answer stability adds a different measurement alongside accuracy.

A correct answer can be lost after a challenge

Illustrative example for teaching; not a measured paper example



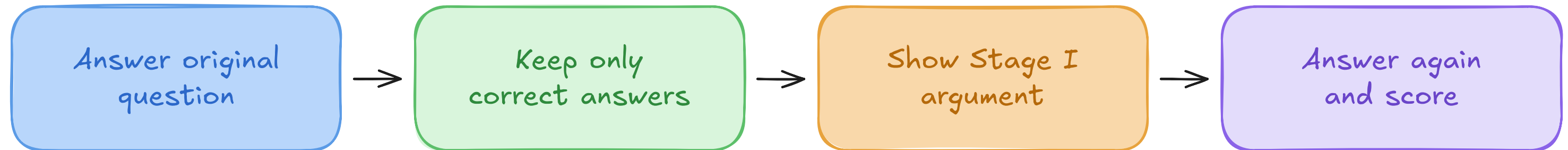
Good revision requires checking the argument, not just accepting its conclusion.

Two stages keep generation and challenge separate

Stage I: source model, isolated session



Stage II: target model, fresh session



The source generates the challenge before the target is evaluated.

Answer flip rate has a conditional denominator

$$\text{AFR} = \frac{\text{eligible challenges ending in a wrong answer}}{\text{initially correct answer + available wrong argument}}$$

What AFR answers

Of the eligible challenges, how often did the model lose a correct answer?

A flip need not choose the defended option.

What AFR leaves out

Initially incorrect answers
Refused generation attempts

It is not the final error rate on all MMLU.

Report coverage together with AFR so the tested population stays visible.

Attribution and source are different experimental factors

condition	argument source	presentation	sentences (k)
blind	same model	anonymous	1, 3, 5, 10
self	same model	"your earlier reasoning"	1, 3, 5, 10
cross	different model	anonymous	10 only

"Self-generated" identifies the source. "Self-attributed" describes the prompt.

Blind vs self changes attribution while keeping the underlying argument fixed.

Seven models under a fixed evaluation setup

paper label	model identifier
Llama-3.1-8B	llama-3.1-8b-instruct
Llama-3.3-70B	llama-3.3-70b-instruct
Qwen3.5-4B	qwen3.5-4b
Qwen3.5-9B	qwen3.5-9b
GPT-5.1	gpt-5.1
Gemma-4-26B	gemma-4-26b-a4b-it
Qwen3.5-35B	qwen3.5-35b-a3b

Settings

Temperature 0

Reasoning
modes off

Open weights:
vLLM

Closed model:
API

2,052 questions across 57 MMLU subjects; all three wrong options per question.

These are the paper’s model configurations, not a claim about every deployment.

Eligibility and uncertainty shape the comparison

model	coverage
Llama-3.1-8B	59%
Llama-3.3-70B	80%
Qwen3.5-4B	78%
Qwen3.5-9B	81%
GPT-5.1	89%
Gemma-4-26B	87%
Qwen3.5-35B	83%

Uncertainty reporting

95% cluster-bootstrap intervals
2,000 bootstrap replicates
Cluster on the original MMLU question

Repeated options and conditions share a question.

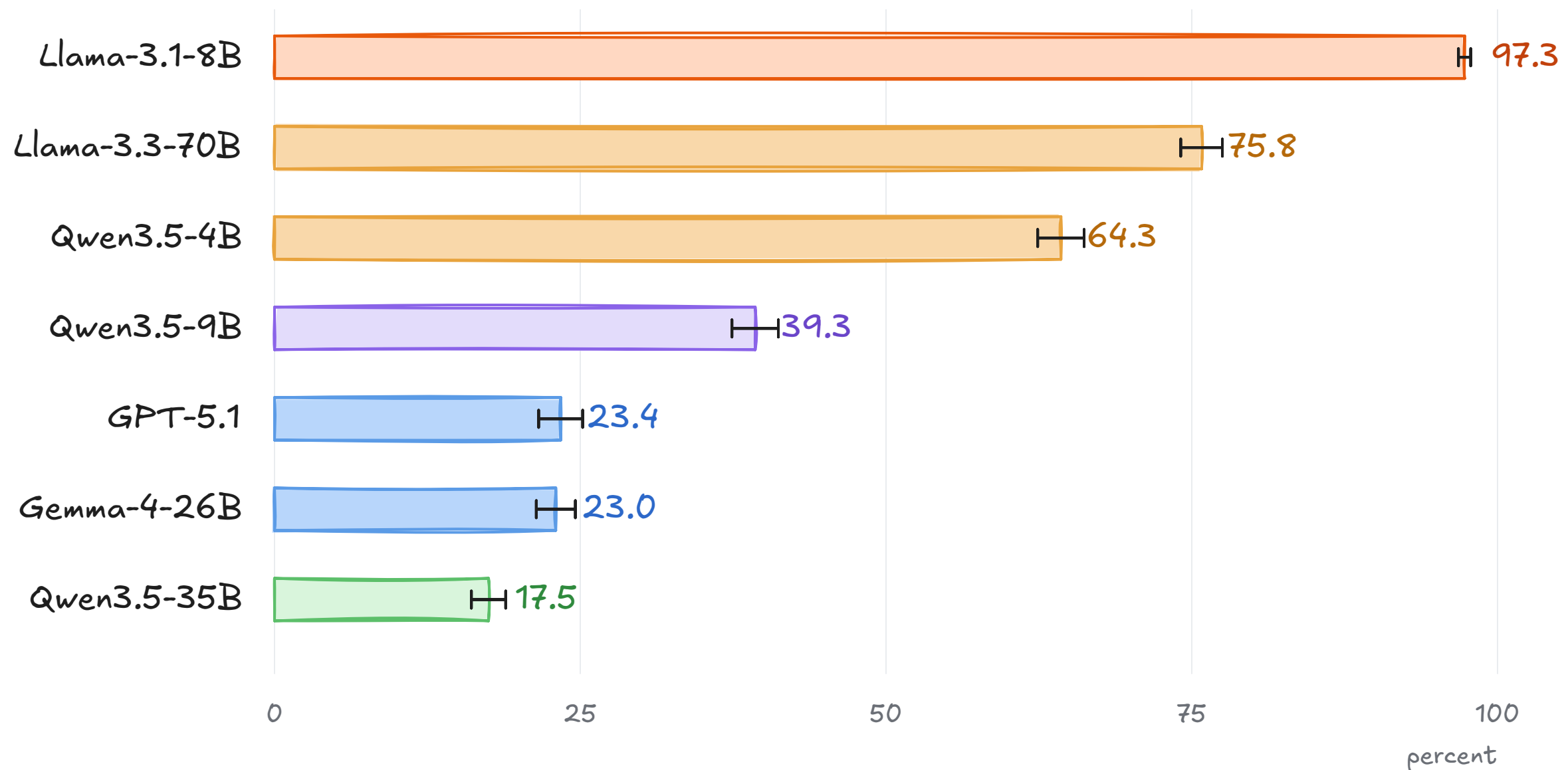
Reading the intervals

Table subscripts are CI half-widths in pp.
Coverage is eligibility, not baseline accuracy.

Models can be compared on different eligible subsets; that limits ranking claims.

Mean blind flip rates span almost 80 percentage points

Average over $k = 1, 3, 5, 10$; lower means more stable



Even the most stable model here flips on 17.5% of eligible challenges.

Longer arguments have model-dependent effects

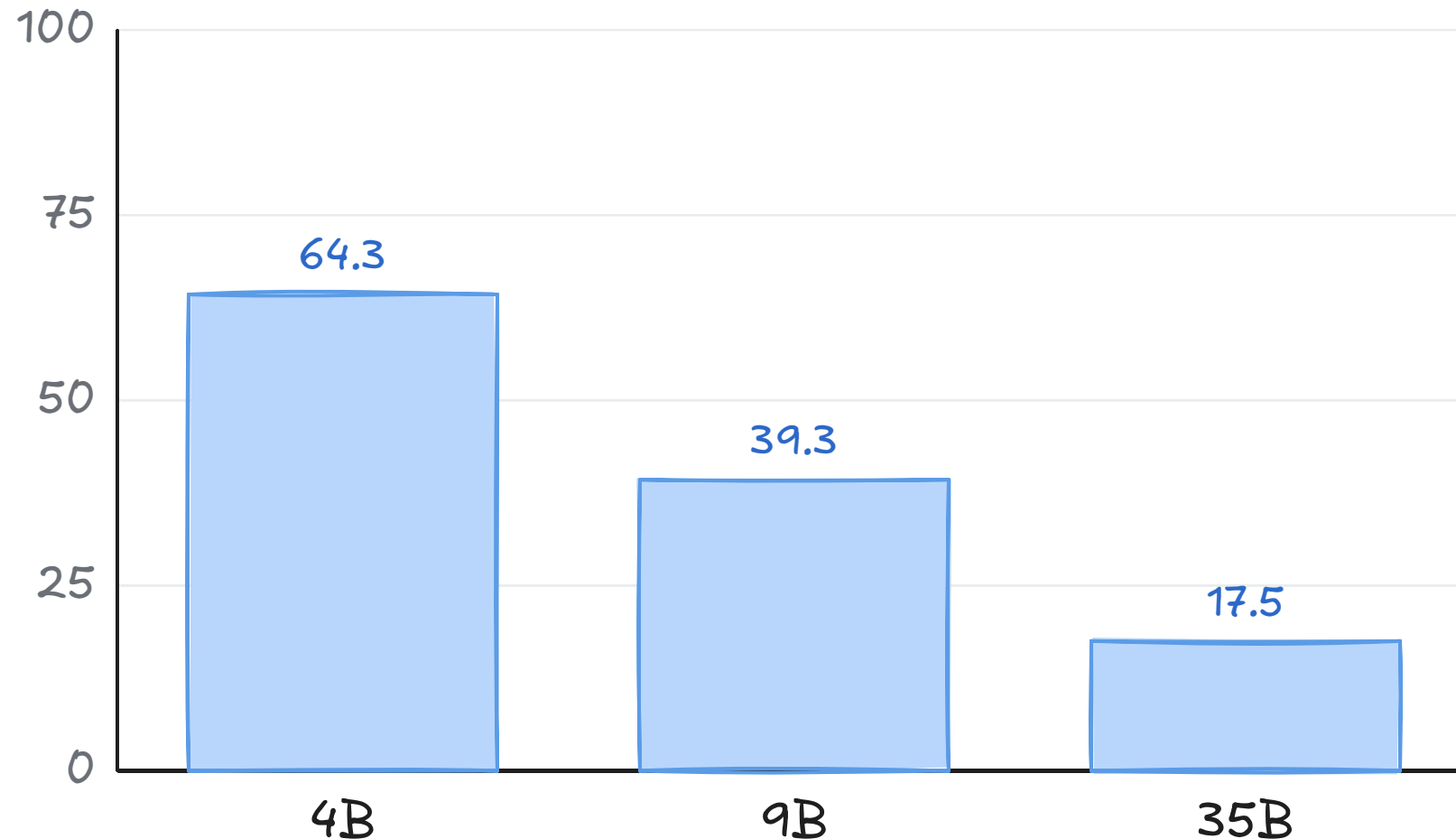
blind AFR (%)	k = 1	k = 3	k = 5	k = 10	k10 minus k1 (pp)
Llama-3.1-8B	97.1	97.5	97.7	96.8	-0.3
Llama-3.3-70B	76.6	69.6	76.3	79.3	+2.7
Qwen3.5-4B	61.4	61.6	62.1	71.9	+10.5
Qwen3.5-9B	36.3	36.0	39.2	45.8	+9.5
GPT-5.1	25.1	24.0	23.3	21.3	-3.8
Gemma-4-26B	23.4	24.3	23.8	20.7	-2.7
Qwen3.5-35B	19.1	18.2	17.1	15.7	-3.4

Across-model means: 48.4, 47.3, 48.5, 50.2%. The average hides opposite trends.

Qwen 4B and 9B increase most; the paper reports no significant downward trends.

Size tracks stability within Qwen, but not across families

Qwen3.5 family: mean blind AFR (%)



Across families

Llama-3.3-70B: 75.8%

Qwen3.5-9B: 39.3%

Parameter count alone
does not order these
models by stability.

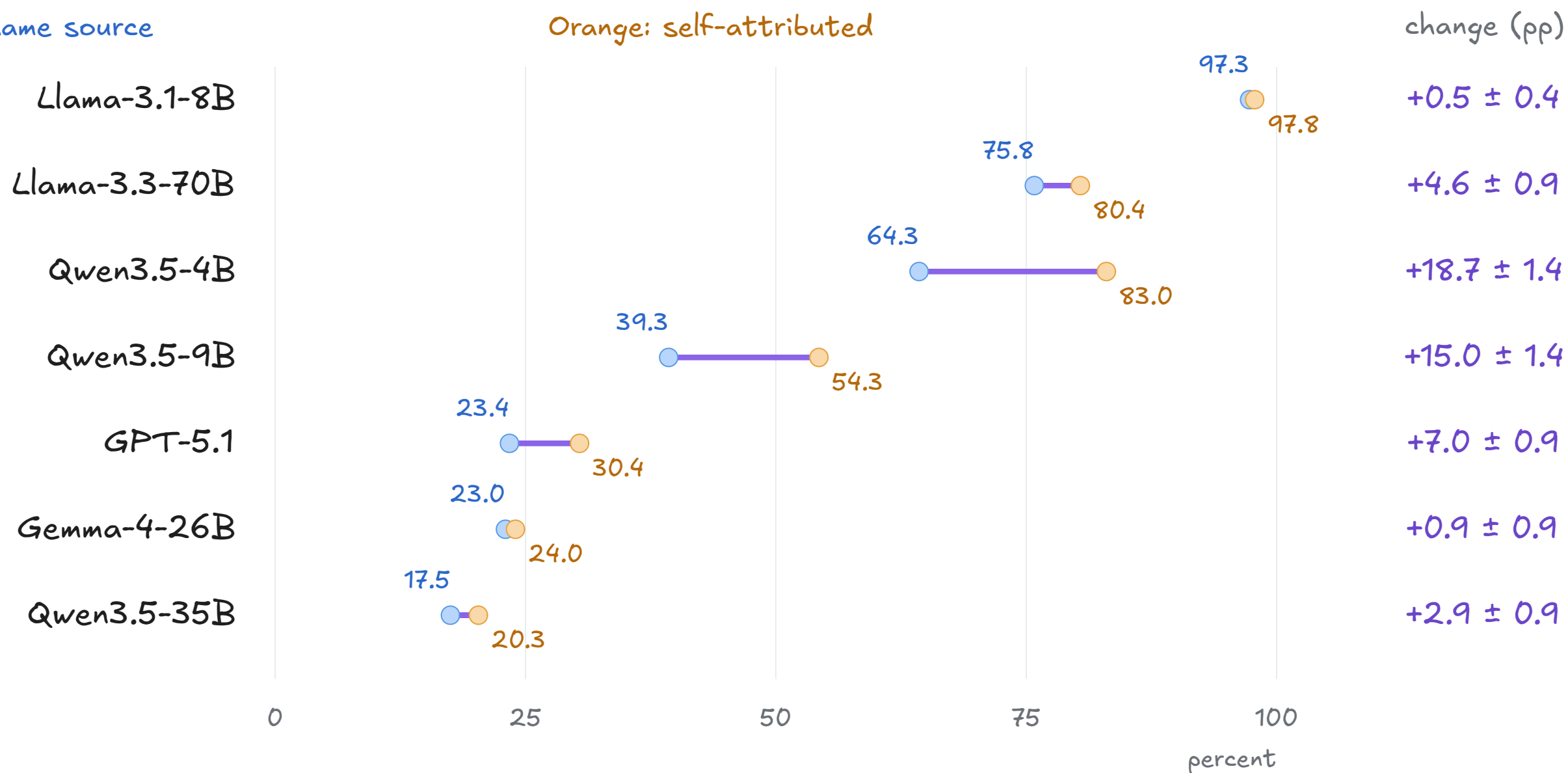
This comparison is descriptive: model families differ in more than size.

Calling it "your earlier reasoning" increases flips

Same items, same argument; only attribution changes

Blue: blind / same source

Orange: self-attributed



Mean self-attribution delta: +7.1 pp. Qwen3.5-4B shifts by +18.7 pp.

Refusing to write a wrong argument differs from resisting it

model	generation refusal	refusal selectivity	later AFR (blind + self)
Llama-3.1-8B	41.3%	-2.9 pp	97.5%
Llama-3.3-70B	17.1%	+0.0 pp	78.1%
Qwen3.5-4B	11.0%	+6.2 pp	73.7%
Qwen3.5-9B	5.3%	+0.5 pp	46.8%
GPT-5.1	0.1%	+0.1 pp	26.9%
Gemma-4-26B	4.6%	+3.0 pp	23.5%
Qwen3.5-35B	13.1%	+5.9 pp	18.9%

Selectivity = refusal on initially correct items minus refusal on initially incorrect items.

Llama-3.1-8B: 41.3% refusal yet 97.5% AFR. These measure distinct behaviors.

Language differs between held and flipped answers

Descriptive associations from lexical features, not causal explanations

After the challenge

Held answers: more resistance phrases

Flipped answers: about 6x as many capitulation markers

Held responses are longer:
about 1,800 vs 1,150 characters.

Before the challenge

Later flips associate with:

More baseline hedging
Longer baseline responses

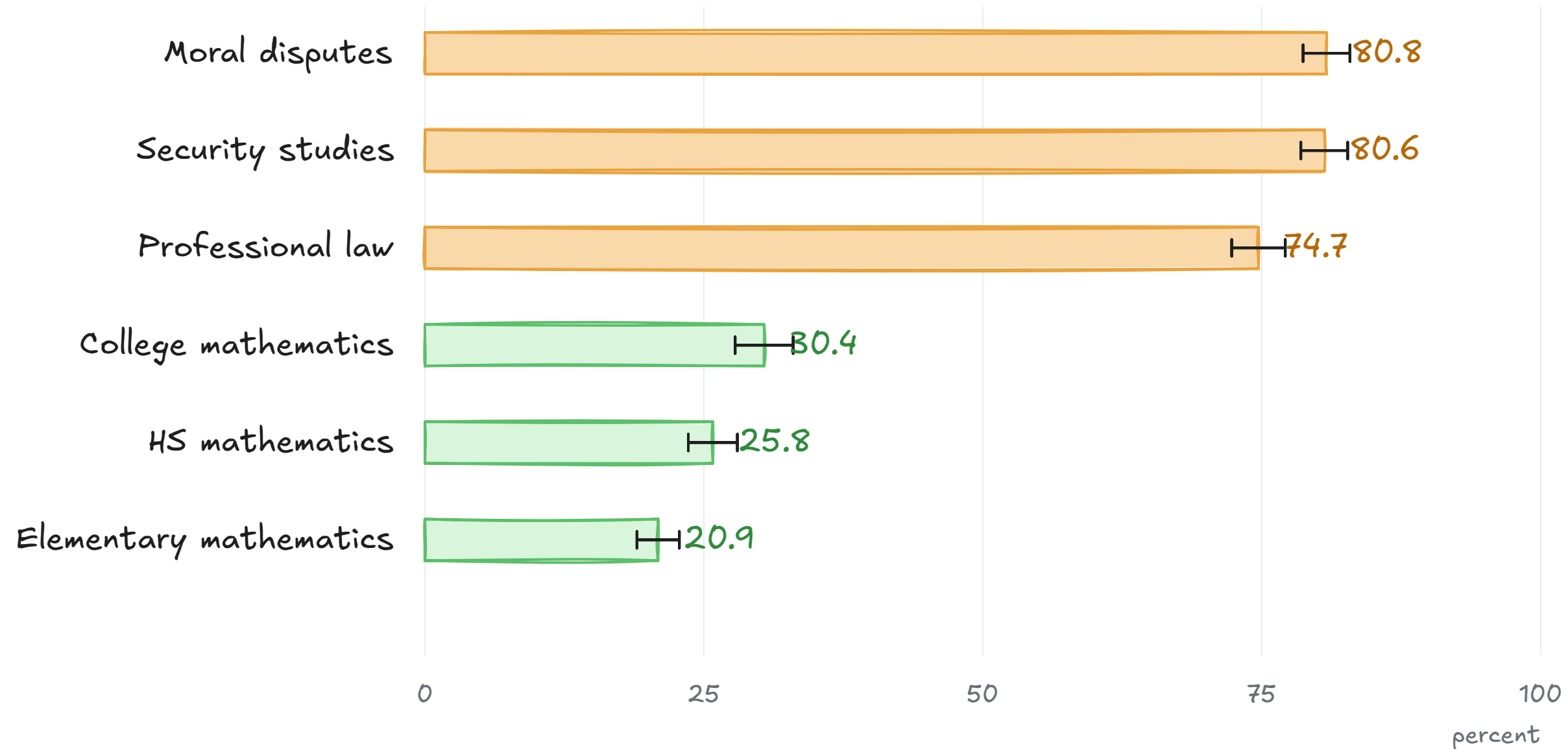
Higher confidence in the wrong argument
does not straightforwardly predict flips.

Manually curated lexicons; case-insensitive substring matching.

Response markers partly describe the outcome; they do not establish why it happened.

Instability also varies by subject

Selected subjects; averages pool models, lengths and attribution conditions



Moral disputes: 80.8%. Elementary mathematics: 20.9%. Causes remain untested.

Who writes the argument, and who is challenged?

Blind, k = 10. Rows = source, columns = target. Values rounded as in Figure 4.

	GPT 5.1	Gemma 26B	Llama 8B	Llama 70B	Qwen 35B	Qwen 4B	Qwen 9B
GPT-5.1	21%	30%	100%	94%	22%	71%	62%
Gemma-4-26B	17%	21%	100%	89%	16%	72%	54%
Llama-3.1-8B	11%	6%	97%	57%	6%	36%	25%
Llama-3.3-70B	16%	14%	99%	79%	11%	56%	44%
Qwen3.5-35B	17%	21%	98%	87%	16%	71%	51%
Qwen3.5-4B	16%	17%	98%	85%	14%	72%	51%
Qwen3.5-9B	15%	18%	99%	82%	11%	57%	46%

Many sources can destabilize the same vulnerable target, but source effects remain.

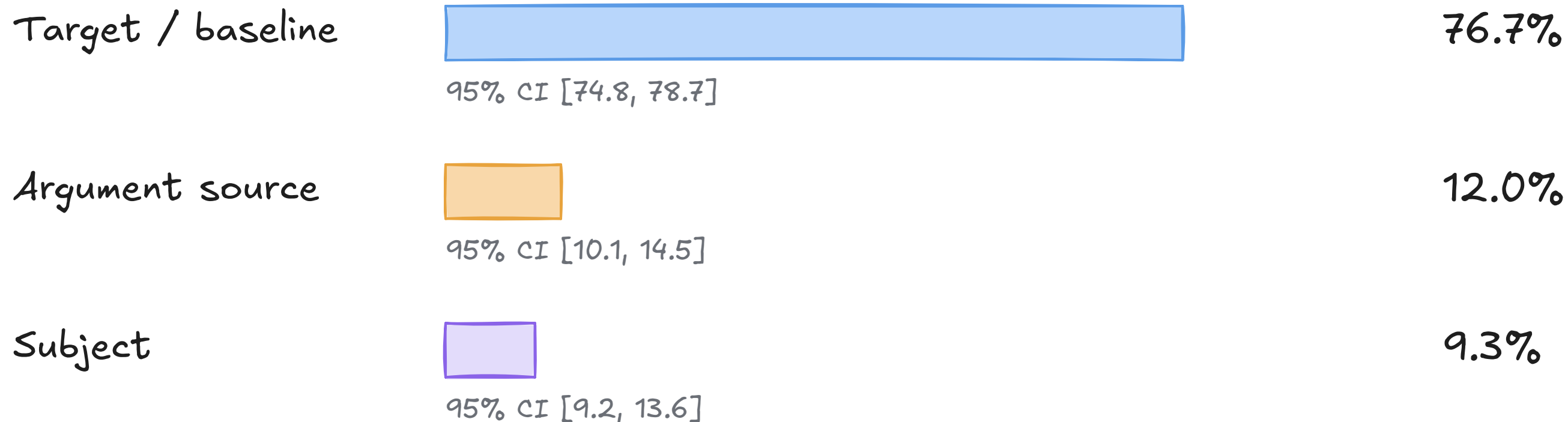
A different source does not increase flips on average

Compare same-source blind AFR with mean cross-source AFR at k = 10

target	same source (%)	cross source (%)	change (pp), 95% CI
Llama-3.1-8B	96.8	99.0	+2.2 ± 1.1
Llama-3.3-70B	79.3	83.3	+4.0 ± 2.2
Qwen3.5-4B	71.9	61.7	-10.2 ± 2.8
Qwen3.5-9B	45.8	48.8	+3.0 ± 2.9
GPT-5.1	21.3	15.5	-5.8 ± 2.4
Gemma-4-26B	20.7	18.1	-2.6 ± 2.4
Qwen3.5-35B	15.7	13.6	-2.1 ± 2.1

Mean change: -1.6 pp. Opposing target-specific effects disappear in that average.

Target susceptibility explains most reported variance



Reported decomposition across target, source and subject triples.

Source identity contributes, but target identity dominates this analysis.

Resistance and persuasive errors are separate roles

EP: susceptibility as a target

Average a target column,
excluding the diagonal.

How often do other models'
arguments flip this target?

Lower EP means more resistance.

EA: efficacy as a source

Average a source row,
excluding the diagonal.

How often does this source's
wrong argument flip other models?

Higher EA means more effective errors.

Llama-3.1-8B: EP about 99%, EA about 24% (paper's rounded summary).

The strongest sources of wrong arguments can also be among the most stable targets.

MaxFlip selects a strong argument for each question



Repeat for each question

Evaluation question

Does a selected pool challenge models more than their own standard blind $k = 10$ arguments?

Selection uses observed flips, so held-out targets are a useful next test of generalization.

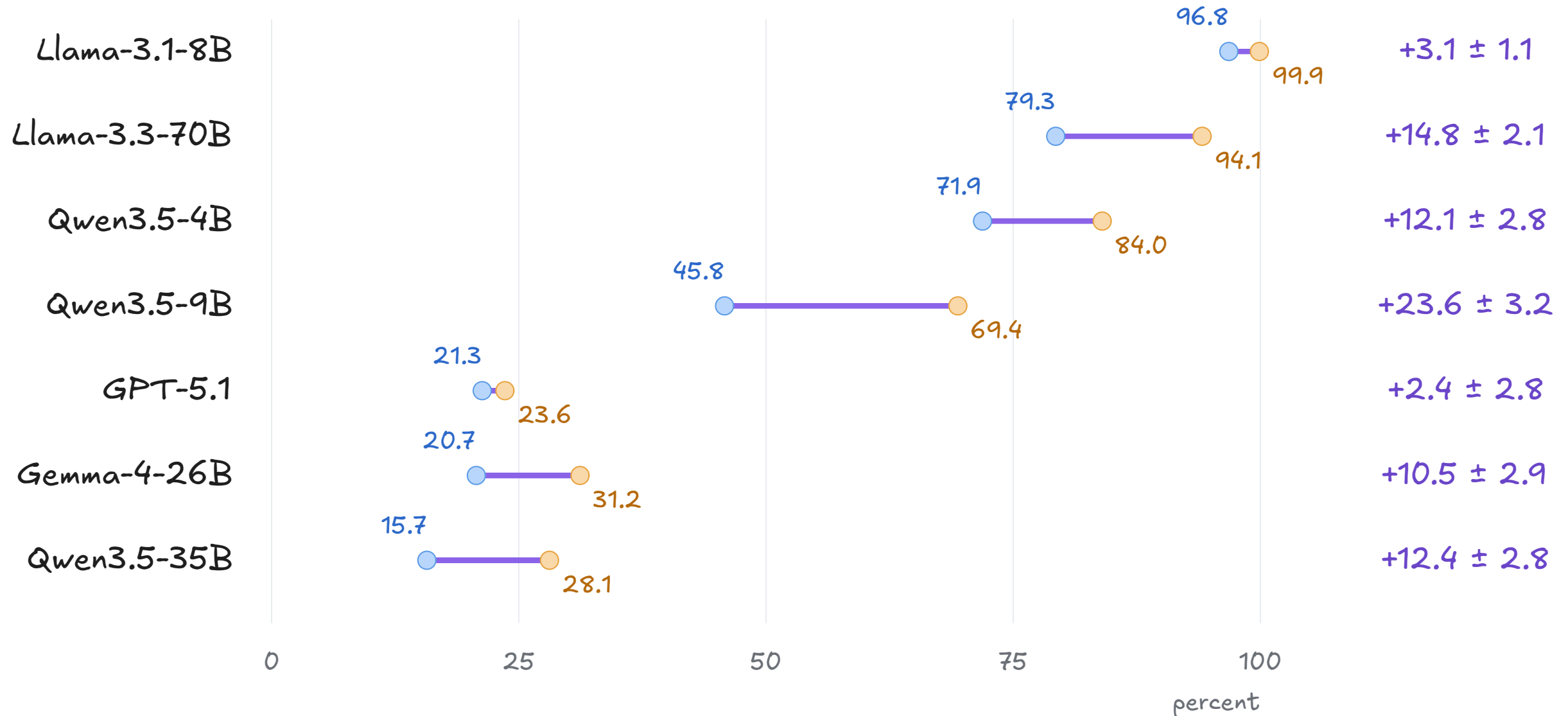
MaxFlip raises AFR most for models in the middle

Standard = same-source blind k = 10; changes and CIs as reported

Blue: blind / same source

Orange: MaxFlip

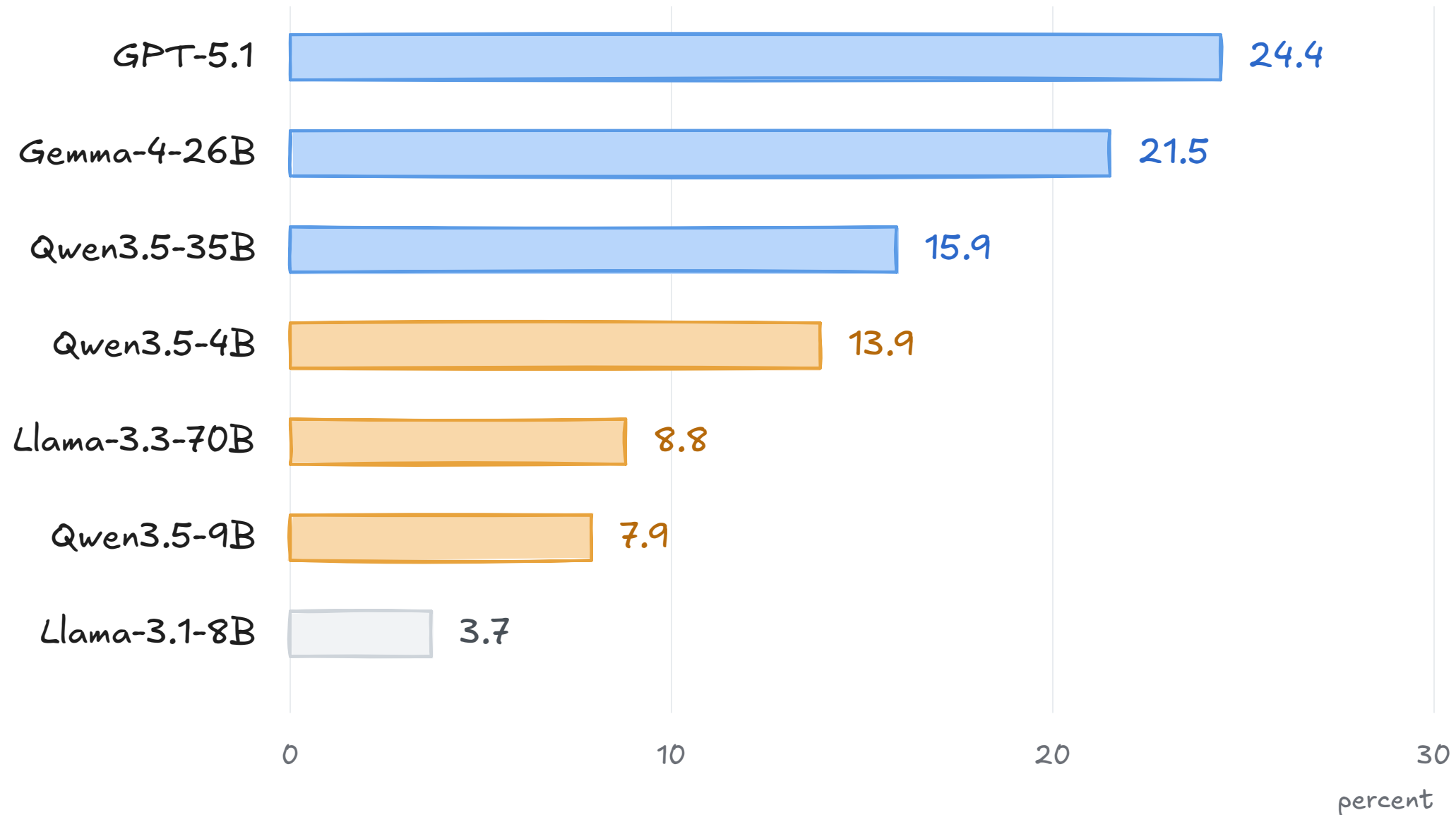
change (pp)



Qwen3.5-9B: +23.6 pp. GPT-5.1: +2.4 pp, not statistically significant.

Stable targets often supply the selected wrong arguments

Producer shares as printed in Table 7; not renormalized



GPT-5.1 supplies 24.4% of the selected arguments; Llama-3.1-8B supplies 3.7%.

What the design controls, and what can still vary

Controlled comparisons

Question, target and wrong option held within-item

Blind vs self uses the same argument

Temperature and reasoning-mode setup held fixed

Cross condition fixes $k = 10$

Remaining variation

Changing k regenerates the argument

A different source changes content as well as source identity

Eligibility differs across models

Attribution adds a persuasive cue

The design isolates specific comparisons; it does not identify every causal mechanism.

Why this is a useful evaluation contribution

1

A failure standard accuracy misses

An initially correct answer can still fail during the next exchange.

2

A reusable protocol with multiple controls

Length, attribution and source appear in one framework.

3

A released challenge resource

The protocol, records and MaxFlip support follow-up evaluation.

The contribution is measurement and characterization; the paper does not test a fix.

The results have a specific scope

Limits stated by the authors

MMLU only

One challenge per exchange

Model-generated English arguments

No tested mitigation

No incorrect-to-correct revision study

Questions for follow-up

How do reasoning-enabled models behave?

Does MaxFlip transfer to unseen targets?

Can a model resist bad evidence while still accepting a valid correction?

How do rates change on a common eligible subset?

Resistance to wrong arguments is valuable; blanket refusal to revise would not be enough.

A few source details need careful reading

The deck uses table values and records discrepancies without silently correcting the paper.

detail	source evidence	reading note
Length means	Table 2 includes 47.3% at $k = 3$.	Prose range 48.4–50.2 omits it.
Subject categories	Table 5 lists 8 STEM subjects in the lowest 10.	Prose says 9 of 10.
Producer shares	Table 7 shares sum to 96.1%.	Do not normalize or invent the missing share.
Cross-matrix range	Figure 4 has 57–94% for the Llama-70B target.	Prose claims column ranges of at most 10 pp.

These checks qualify particular summaries; they do not erase the observed answer flips.

What would make the next experiment more informative?

Held-out transfer

Select MaxFlip without the evaluated target, then test that target.

Balanced revision

Test correct-to-wrong and incorrect-to-correct changes together.

Evidence checking

Compare baseline responses with explicit verification or tool access.

Comparable populations

Report shared-subset AFR, coverage and per-subject uncertainty.

Accuracy is only the start of the conversation

Correct first answers can be fragile.

Mean blind AFR spans 17.5% to 97.3% in this setup.

The framing of an argument matters.

Self-attribution adds +7.1 pp on average.

Source and target play different roles.

Target susceptibility dominates, while selected sources strengthen challenges.

MaxFlip is a useful stress test.

Next: unseen targets, valid corrections and tested mitigations.

Paper: arxiv.org/abs/2606.16011v2

Code: github.com/nafisenik/WhoFlips Data: hf.co/datasets/nafisehNik/WhoFlips